

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. (CL) Examination (Old Course)

November 2013

CL302 Structural Analysis-II

Date: 20.11.2013, Wednesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

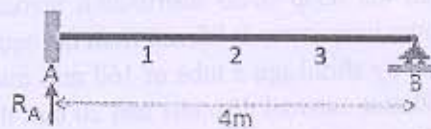
SECTION - I

Q - 1 A cable loaded with 10 kN/m is stretched between two supports in the same horizontal line 200m apart. If the central dip is 15m, find the maximum and minimum pulls in the cables. [07]

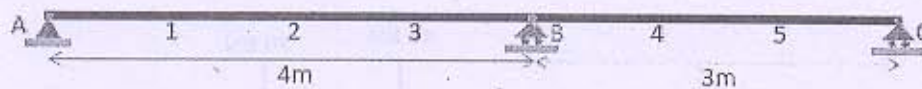
Q - 2 (a) A cable supported at its ends 40m apart carries loads of 20 kN, 10 kN and 12 kN at distances 10 m, 20 m and 30 m from the left end. If the point on the cable where the 10 kN load is supported is 13 m below the level of the end supports. Determine the reactions at the supports. [04]

(b) Attempt any One. [10]

(i) Sketch ILD for R_A , R_B , M_A giving ordinates at 1m intervals for the propped cantilever shown in the Figure.



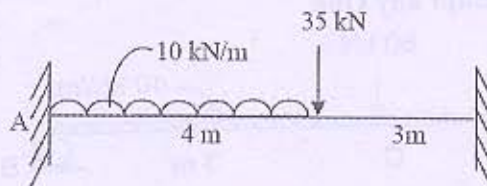
(ii) Sketch ILD for M_B giving ordinates at 1m intervals for the two span continuous beam shown in the Figure. Assume EI is constant throughout.



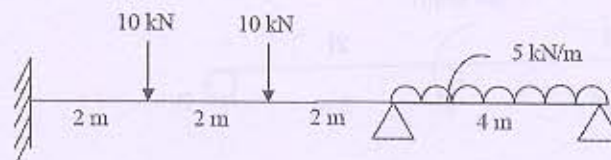
Q - 3 (a) Why influence line diagram for Statically Indeterminate Structure is a curve? [02]

(b) Find out support reactions and draw shear force and bending moment diagrams. (Attempt any One) [12]

(i)



(ii)

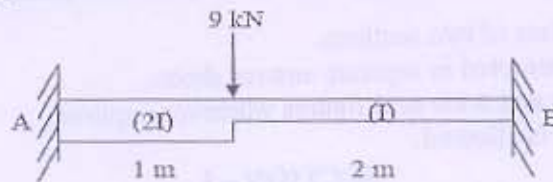


SECTION - II

Q - 4 Determine the strain energy of a cantilever beam of span 2 m having size 20 mm width x 60 mm depth. Take $E = 200 \text{ GPa}$. [07]

- (a) When 1000 N concentrated load is placed at free end.
 (b) When total 1000 N load is uniformly distributed over the entire span.

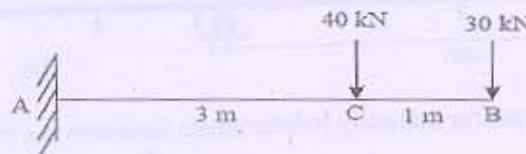
Q - 5 (a) Calculate fixed end moments for a beam loaded as shown in Figure using theorem of minimum strain energy. [04]



(b) Attempt any Two [10]

- (i) A cast iron pipe of 400 mm internal diameter and 100 mm thickness carries water under a pressure of 8 kN/mm^2 . Determine the maximum and minimum intensities of hoop stress across the section. Also sketch the radial pressure distribution and hoop stress distribution across the section.
- (ii) A thick cylindrical pipe of internal radius of 12 cm and external radius 16 cm is subjected to an internal fluid pressure of 120 kg/cm^2 . Determine the maximum and minimum hoop stresses in the cross-section. Also sketch the hoop stress distribution across the section. What is the percentage error if the maximum hoop stress is found from the equation for thin pipes.
- (iii) A compound cylinder is made by shrinkage a tube of 160 mm internal diameter and 20 mm thick over another tube of 160 mm external diameter and 20 mm thick. The radial pressure at the common surface, after shrinkage, is 8 N/mm^2 . Find the final stresses set up across the section, when the compound cylinder is subjected to an internal fluid pressure of 60 N/mm^2 .

Q - 6 (a) Using energy principles, calculate θ_B and δ_B for a beam shown in Figure. Take $E = 2 \times 10^4 \text{ N/mm}^2$, $I = 5 \times 10^8 \text{ mm}^4$. [05]



(b) Analyze the beam shown in Figures by consistent deformation method and draw S.F. and B.M. diagrams. **Attempt any One.** [09]

